When Splits Matter: Interpreting Disaster Tweet Classification Models

1st Jamie Livers
Department of Computer Science
Bowling Green State University
Bowling Green, Ohio, USA
jlivers@bgsu.edu

2nd Nicholas Johnson
KDWM Lab
University of Louisville
Louisville, Kentucky, USA
nicholas.johnson.8@louisville.edu

3rd Kyle Spurlock
KDWM Lab
University of Louisville
Louisville, Kentucky, USA
kyle.spurlock@louisville.edu

4th Olfa Nasraoui
KDWM Lab
University of Louisville
Louisville, Kentucky, USA
olfa.nasraoui@louisville.edu

2025 IEEE International Conference on Data Mining Workshops (ICDMW) | 979-8-3315-8132-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDMW69685.2025.00363

***Abstract*—Classification models are increasingly used to classify disaster-related tweets, supporting real-time decision-making during crises. However, common evaluation practices, such as random data splits, can inflate performance by allowing disaster-specific vocabulary to leak between training and testing sets. We propose a new evaluation framework for disaster event classification, called Disaster-Holdout Validation (DHV), extending the Leave-One-Disaster-Out (LODO) strategy used in previous work. DHV offers a more realistic alternative by simulating unseen disaster scenarios. This study examines how data splitting strategies affect not only classification performance but also model interpretability. Using a labeled dataset of disaster tweets, we train linear support vector machines (SVMs) on four tasks: informativeness, first responder relevance, disaster type, and secondary annotation. We compare SHAP (SHapley Additive exPlanations) values and SVM coefficients to analyze feature importance and use Kendall's Tau to measure ranking consistency across splits. Our results show that DHV-trained models emphasize more context-specific and generalizable features, while random splits favor generic or high-frequency terms. To support experimentation, we developed a graphical interface for running and visualizing model outputs. Overall, DHV better reflects real-world generalization, although at some cost to feature stability, highlighting the trade-offs between performance and explainability in crisis informatics.**

***Index Terms*—disaster response, explainable AI, Disaster-Holdout Validation (DHV), Leave-One-Disaster-Out (LODO), tweet classification, data splitting, SVM, feature attribution, crisis informatics Introduction**

I. Introduction

Machine learning models have become increasingly central to real-time disaster response systems, particularly in classifying and processing social media data like tweets. These models are used to filter relevant information during emergency crises, identify urgent needs, and help support decision-making for first responders and humanitarian organizations. However, the reliability of these models is often shaped by the data splitting strategy used during training and evaluation. Although most models are evaluated using standard random shuffling, this approach can produce overly optimistic results, particularly when disaster events differ significantly in language, context, or scale.

Disaster tweet data are difficult to model because vocabulary changes across events, context is highly specific, and class labels are often uneven. For example, language patterns during a wildfire differ significantly from those during an earthquake or a flood. Evaluating models with random splits allows tweets from the same event to appear in both training and testing sets, making it difficult to assess how well a model generalizes to unseen disasters. A more robust approach involves Disaster Holdout Validation (DHV), in which models are trained on all disasters except one and then tested exclusively on the held-out event. This simulates real-world deployment conditions, where models must interpret tweets from a new disaster they have never seen before.

In this study, we investigate how these data-splitting strategies affect not just model performance, but also the features the model learns to make predictions. Using a linear Support Vector Machine (SVM) and a disaster tweet dataset, we compare random and DHV splits across four classification tasks: informativeness, first responder relevance, disaster type, and secondary annotations. We apply SHAP (SHapley Additive exPlanations) and SVM coefficient analysis to interpret the learned features, and we further assess the consistency of the feature ranking using Kendall Tau.

To support experimentation, we also developed a graphical user interface (GUI) that allows researchers to select disasters and tasks, perform individual or batch experiments, and export results for analysis. Our goal is to determine whether DHV yields meaningfully different or more event-specific feature importance compared to traditional random splits and what implications this has for model interpretability and disaster response reliability.

II. Background

Machine learning methods for disaster-related tweet classification have advanced significantly. In particular, SentiBERT-

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BiLSTM-CNN has demonstrated improved performance on classification tasks by combining sentiment-aware embeddings with sequential and convolutional layers, highlighting the importance of robust feature representations in this domain [1].

To reflect real-world deployment, researchers have adopted evaluation frameworks like Leave-One-Disaster-Out (LODO) or Leave-One-Group-Out (LOGO), which simulate domain shift by excluding entire disaster events from training sets, unlike random splitting, which can lead to inflated performance metrics due to vocabulary overlap [2]. Previous work has also applied this approach directly to disaster tweet classification: Li et al. (2021) combined self-training with CNN and BERT/BERTweet models under a leave-one-disaster-out framework, showing that self-training improves performance when large amounts of unlabeled target data are available, but offers limited benefit otherwise [3].

The field of Explainable AI (XAI) has also made substantial strides. Within this field, SHAP stands out as a widely adopted method for its consistency and local-global interpretability in text-based models [4]. Understanding explanatory consistency under data distribution shifts is critical. Several recent studies have evaluated XAI metrics and explanation quality using quantitative benchmarks and human validation, emphasizing the need for reliable evaluation frameworks [5], [6]. Furthermore, the use of Kendall Tau to compare feature rankings has emerged as a robust approach to measuring the stability of explanations between experimental variations, including different data splits [7].

Beyond social media classification, machine learning has also been applied to broader disaster resilience challenges. Fatima et al. [8] provide an extensive review of ML techniques for forecasting power outages during hurricanes, focusing on data sources, preprocessing methods, and algorithmic strategies to support infrastructure resilience. Their survey highlights the broader importance of evaluation design in disaster-focused ML systems. Similarly in 2023, the GAO [9] issued a technology assessment of AI in natural hazards modeling, identifying opportunities, such as improved forecasts, ensemble modeling, and synthetic data, as well as challenges, including trust, data gaps, and bias.

Our study builds on these foundations by combining DHV evaluation, SHAP explanations, linear SVM models, and Kendall's Tau comparisons to pinpoint how different data splitting strategies influence the interpretability of disaster tweet classification.

III. Methodology

A. Dataset

This study uses a labeled dataset of disaster-related tweets, originally collected from Twitter (now X) during various natural disasters. Each tweet is annotated for four different classification tasks:

- Informativeness: whether a tweet contains useful disaster information
- First Responder Relevance: whether the tweet is actionable for emergency services
- Disaster Type: the kind of disaster involved (e.g., earthquake, flood, wildfire)
- Secondary Annotation: auxiliary information (e.g., infrastructure damage, missing persons)

The First Responder Relevance task shows a notable class imbalance, with the majority of samples labeled as Other. This reflects the real-world frequency of tweets that do not explicitly reference a specific responder group. While this imbalance may influence model performance toward the majority class, our focus on interpretability and comparison across split strategies (rather than raw accuracy) mitigates its impact on the study's main findings.

B. Data Splitting Strategies

We compare two data-splitting strategies, and each classification task is evaluated separately for both strategies. Each model is trained independently for a single classification task and evaluated on its respective test set.

- Random Split: The entire dataset is shuffled and divided into training (80%) and testing (20%) subsets without regard to disaster boundaries. This is a standard practice in many ML applications but may leak event-specific context between splits, leading to overly optimistic performance estimates.
- DHV: For each run, one disaster event is completely held out as the test set, while models are trained on all remaining events. This setup simulates real-world deployment conditions where a model encounters tweets from a previously unseen disaster, enabling evaluation of its cross-event generalization ability.

SHAP values and other interpretability analyses are computed only on the test-set instances for each split type, ensuring that feature importance reflects model behavior under true evaluation conditions. Our definition of DHV is equivalent to the leave-one-disaster-out strategy used in prior work (e.g., Li et al. [3]), but here we focus on its impact not only on performance but also on interpretability.

C. Model Architecture

We use a linear Support Vector Machine (SVM) for classification due to its strong generalization power on high-dimensional, sparse text data and its transparent interpretability. Since the focus of this study is understanding how different data-split strategies affect feature attribution and generalization, SVM provides a consistent and explainable baseline.

Each classification task (informativeness, first responder relevance, disaster type, and secondary annotation) is trained independently using default hyperparameters unless otherwise specified. Input features are represented as TF-IDF vectors, and the linear kernel allows direct access to feature weights (coefficients), which are later compared with SHAP-derived feature importances.

Although deep neural models such as BERT-based architectures and SentiBERT BiLSTM-CNN have demonstrated

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strong performance in prior work (e.g., Li et al., 2021), this study emphasizes generalization and explainability rather than maximizing predictive accuracy, making SVM a more suitable choice.

D. Explainability and Feature Analysis

To interpret what features the model uses for prediction, we use two complementary approaches.

- SVM coefficients: Since we use a linear kernel, the learned weights for each word feature indicate its importance. The highest positive and negative coefficients are extracted for each class in each task.
- SHAP values: We apply SHAP to compute the contribution of each feature to individual predictions. We aggregate SHAP values across samples to identify the 15 most influential features per class and task.

SHAP values are computed only on the test-set instances for each split configuration. This ensures that the explanations reflect model behavior under true evaluation conditions. Because SHAP features are derived solely from the test data, the space of terms identified may differ between Random and DHV settings—particularly when certain event-specific terms appear only in one subset. This distinction helps explain why feature rankings vary across split strategies.

To assess the stability and consistency of feature importance across splitting strategies, we computed the Kendall rank correlation coefficient (τ) between the ranked lists of features for the random and DHV models. A high τ score indicates that the model relies on similar features regardless of the split method, while a low score suggests that different features are learned under different data regimes.

E. Custom Experimentation Interface

To streamline experimentation, we developed a custom Python-based GUI, shown in Fig. 1. This interface supports reproducibility, reduces configuration overhead, and automates the generation and organization of analysis results. The tool supports:

- Disaster and task selection
- Analysis modes (single run, full batch across tasks or disasters)
- Three preset quality modes for fast, balanced, or high-detail SHAP analyses
- Custom parameter input, including SHAP sample size, number of features, background set, and TF-IDF settings such as max features, n-gram size, and document frequency
- Save model outputs, SHAP plots, and top features results by disaster event and classification task.

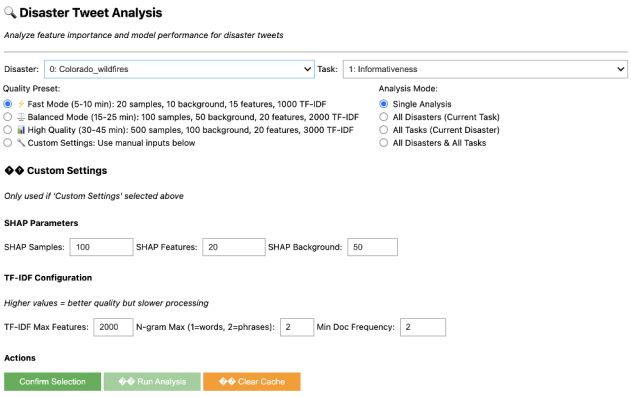


Fig. 1. GUI Interface

IV. Experimental Results

To evaluate how data splitting strategies affect the interpretability of machine learning models for disaster-related tweet classification, we conducted experiments across the four classification tasks listed in Sec. III-A. However, in this paper, we will present the results of two representative tasks, Informativeness (Task 1) and First Responder Relevance (Task 2), while limiting the analysis to four disaster events: Guatemala Earthquake, Philippines Floods, Spain Train Crash, and West Texas Explosion. Each task was evaluated using both DHV and random split strategies. All models were trained using SVM on TF-IDF features. Feature explanations were derived from SHAP values and SVM coefficients.

A. Model Performance

Although model accuracy was not the focus of this study, performance metrics help to confirm that both training strategies produced viable classifiers. For example, in Task 1 (Informativeness) on the Guatemala earthquake, we obtained the following metrics:

- **Random Split** – Accuracy: 0.7909, F1: 0.7846
- **DHV** – Accuracy: 0.7581, F1: 0.7622

This pattern, higher accuracy for random splits, was consistent across other disasters and tasks, as expected, due to the absence of domain shift in random splitting.

B. SHAP-Based Feature Attribution

To understand how feature importance differs between data splits, we generated bar plots of the top SHAP-ranked features for each disaster-task combination. Using a human-annotated relevance scheme, we compared the interpretability of features surfaced by DHV and random split models. In the Guatemala earthquake example (Fig. 2), the DHV model highlights context-aware terms such as "sismo", "muertos", and "terremoto"[1], many of which are labeled as "related and informative." In contrast, the random-split model prioritizes tokens like *"guatemala"* and *"en guatemala,"* which are frequently marked as "not informative" or "not related," suggesting potential overfitting to event-specific vocabulary.

A similar pattern emerges in the Philippines Floods example (Fig. 3), where DHV models emphasize high-utility emergency terms like *"reliefph", "rescueph,"* and *"hospital"*, while the random-split model elevates common disaster terms and

[1]Spanish translations for: sismo and terremoto: earthquake; muerto: dead

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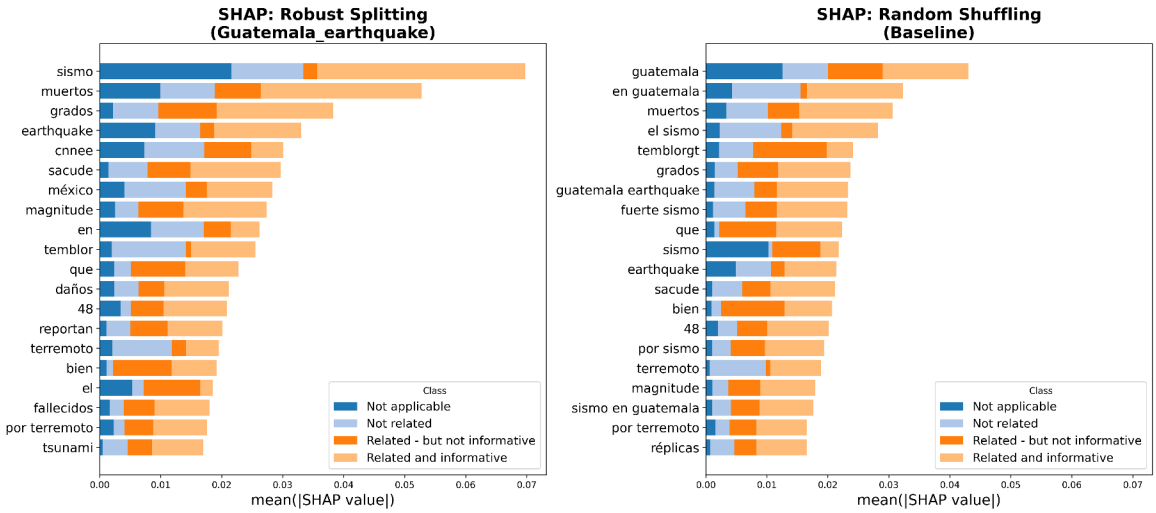


Fig. 2. SHAP feature importance for Guatemala Earthquake (Task 1). The left plot (DHV) emphasizes earthquake-related terms like sismo and muertos. The right plot (random-split) reflects overfitting to localized phrases like *"guatemala"* and *"sismo en guatemala"*, highlighting domain shift in random splits.

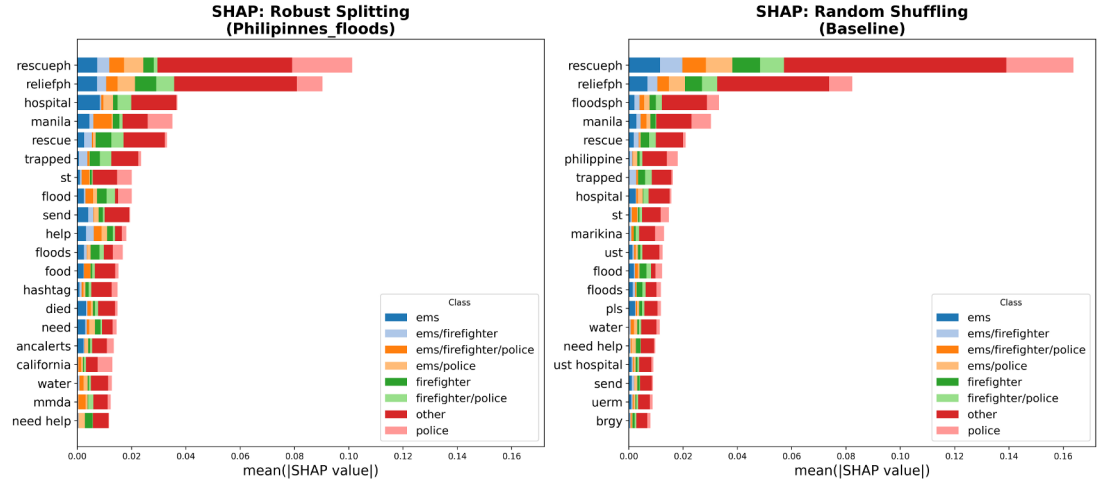


Fig. 3. Top SHAP-ranked features for the Philippines Floods task under DHV (left) and random-split (right) splitting strategies. While both models prioritize shared high-importance terms such as *"rescueph," "reliefph,"* and *"manila,"* the random-split model gives relatively higher prominence to more general aid-related terms like *"help," "need help,"* and *"send."* In contrast, the DHV model places slightly greater emphasis on disaster-specific or institutional terms such as *"mmda"* and *"ancalerts,"* suggesting more context-aware feature attribution.

location-specific tags (e.g., *"marikina", "ust", "brgy"*) that may be less transferable across unseen events. Together, these bar plots reveal that random-split models often favor generic or overfit terms, while DHV models prioritize features more aligned with generalizable, context-sensitive indicators of first responder relevance.

Kendall's Tau analysis supports this distinction, revealing a very low rank correlation (τ = 0.003) between the top features learned under random and DHV splitting strategies (Fig. 4). This sharp divergence shows how data partitioning can significantly influence the linguistic patterns emphasized, potentially affecting a model's ability to generalize in real-world deployments.

SHAP scatter and bar plot comparisons for Task 2 (First Responder Relevance), shown in Fig. 5, reveal substantial divergence in feature importance between random and DHV splits. The scatter plot illustrates that feature attribution rankings are often inconsistent between the two training regimes, with many features influential in the random model receiving low importance scores in the DHV counterpart, and vice versa. This discrepancy suggests that the random model may be overfitting to event-specific tokens, such as names, hashtags, or repeated patterns present in both training and test sets due to data leakage. The DHV model, in contrast, demonstrates more distributed and context-sensitive feature attributions, better aligned with generalizable indicators of first-responder

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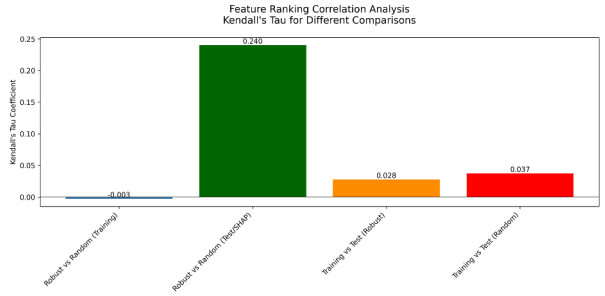


Fig. 4. Kendall's Tau coefficients comparing ranked feature lists across data splits and evaluation phases. The low correlation between training and test rankings under random and robust splitting suggests that models learn different features when generalization is required.

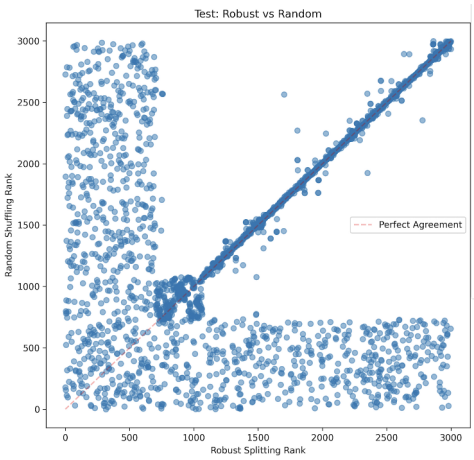


Fig. 5. SHAP rank correlation for Task 2 (First Responder Relevance) on the West Texas Explosion. The plot highlights high agreement between DHV and random-split models for low-importance features, suggesting both models consistently assign low ranks to similar low-signal tokens.

relevance.

The rank correlation pattern in the SHAP scatter plot also provides additional insight: disagreement is more pronounced among the top 1000 ranked features, those with the highest relevance, while convergence appears for less informative tokens ranked below this threshold. This tail agreement suggests that, while both models identify unimportant features similarly, they diverge in identifying the most meaningful ones, especially under data leakage conditions.

The SHAP bar plot confirms this distinction: the random split model emphasizes location-specific or event-memorized tokens like *"plant," "waco," "texas,"* and *"fertilizer,"* while the DHV model highlights broader response-related terms such as *"injured," "dead," "firefighters,"* and *"explosion."* These differences confirm the interpretability advantages of DHV-trained models in real-world disaster response scenarios (Fig. 6). The empirical patterns revealed in these results support the argument that DHV-trained models offer stronger generalization and are better equipped for deployment in real-world disaster response systems, where future events may differ significantly from training scenarios.

C. Rank Correlation Analysis

To quantitatively assess how learned feature importance varies with the data split strategy, we computed Kendall's Tau (τ) between the top 15 features across both splits. The following correlation values indicate the extent to which different splitting methods alter the way the top features are ranked by the model:

- **Guatemala Earthquake – Task 1:** $\tau = 0.53$ (moderate consistency)
- **Philippines Floods – Task 2:** $\tau = 0.32$ (low consistency)
- **Spain Train Crash – Task 1:** $\tau = 0.44$ (moderate-low consistency)
- **West Texas Explosion – Task 2:** $\tau = 0.66$ (high consistency)

These findings support the claim that random splits can lead to overemphasis on shallow patterns, while DHV helps surface features with more semantic or event-specific value.

To further enhance interpretability, we examined individual predictions using SHAP force plots generated from the DHV model. These visualizations reveal how specific tokens, such as hashtags, named locations, or emotionally charged terms, influenced the model's decisions. Focusing on DHV predictions can reveal how the model prioritizes more generalizable, context-aware features instead of relying on event-specific artifacts.

Alongside each force plot, we also include a representative tweet sample showing the original tweet text, the true label, the model's predicted label, its prediction confidence, and the top five SHAP-attributed words driving the prediction. These local explanations offer concrete examples of how the model reasons through difficult cases and demonstrate the interpretive benefits of robust data splitting strategies. These local explanations are presented in the appendix.

V. Discussion

Our preliminary results indicate that the choice of data splitting strategy has a substantial impact on model behavior, particularly in the types of features a model learns to prioritize when classifying disaster-related tweets. Although general accuracy metrics were not the main focus of this study, our SHAP and coefficient analyses suggest that models trained with DHV exhibit greater sensitivity to event-specific language, whereas models trained using random splits tend to learn more generic features that may not generalize as well in deployment scenarios.

This finding supports the idea that random shuffling can mask distributional shifts present in real-world disaster data. When training and testing data come from the same disaster or from events with similar characteristics, the model is more likely to pick up on shallow or surface-level patterns. In contrast, DHV forces the model to generalize across events, encouraging it to identify more semantically meaningful or domain-relevant signals, but sometimes at the cost of performance or stability in feature attribution. Li et al. [3] similarly

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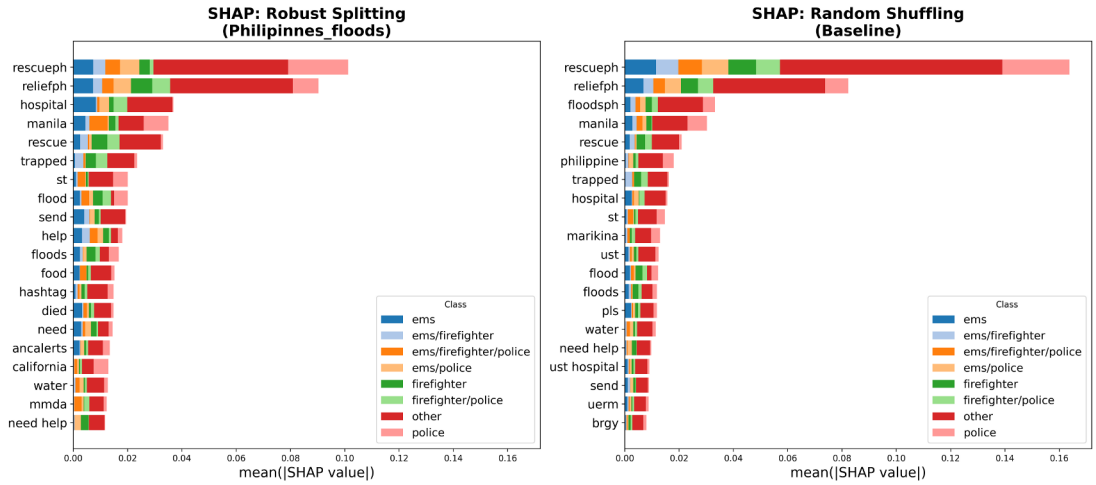


Fig. 6. SHAP feature importance visualizations for West Texas Explosion – Task 2. The left panel shows the top-ranked features by SHAP value under DHV splitting, with a broader distribution of response-relevant terms across classes. The right panel displays the same for the random split baseline, where high-ranking tokens are largely location-specific and repeated phrases from both training and test sets, suggesting memorization and data leakage. Class coloring reveals how term importance varies across first responder categories.

noted performance trade-offs when evaluating deep learning models under a leave-one-disaster-out framework, finding that self-training improved generalization only when sufficient unlabeled target data were available. Our results complement theirs by highlighting that the evaluation strategy not only affects performance but also shifts the interpretability of model features.

Our use of Kendall's Tau to compare feature rankings between splits revealed that feature consistency varies by task, with higher agreement on tasks such as disaster type and lower agreement on tasks such as first responder relevance. This may be due to the fact that some tasks rely more heavily on stable semantic categories (e.g., type of disaster), while others are more context- or culturally dependent (e.g., what counts as useful to a responder in a given region).

These results have implications for both model evaluation and interpretability in crisis informatics. Researchers and practitioners should be cautious when interpreting explanations derived from models trained on randomly split data, as these may not reflect the decision boundaries the model would use in real-world conditions. In high-stakes applications, DHV or group-aware validation may be more appropriate not just for measuring performance, but also for understanding how and why a model makes decisions.

Finally, our GUI framework proved valuable for iterating over experiments and organizing the results. In future work, this tool could be extended to support model comparison or interactive explanation exploration.

VI. Conclusion

This study investigated how different data splitting strategies affect the features learned by machine learning models for disaster-related tweet classification. Using SVMs, we compared standard random splits with a DHV approach, analyzing feature importance through SHAP values and model coefficients. Across four classification tasks, identifying informativeness, first responder relevance, disaster type, and secondary annotation, we observed that the choice of splitting strategy significantly influenced the model's learned representations.

In particular, DHV-trained models tended to focus on more contextually grounded and generalizable terms, while random split models often prioritized dataset-specific patterns. This is likely due to vocabulary leakage, where the same event-specific words (such as location names or hashtags) appear in both training and testing sets. In such cases, models can effectively "memorize" certain terms as shortcuts for classification, inflating performance metrics without learning robust features. The SHAP visualizations clearly highlighted these differences, offering insight into the behavior of both local and global models. Feature ranking comparisons using Kendall's Tau further quantified changes in feature importance, with variability between tasks.

To support experimentation, we developed a flexible GUI for configuring runs, adjusting hyperparameters, and exporting visual and tabular outputs. This interface facilitated efficient analysis and ensured reproducibility by organizing the results by task and event.

These findings highlight the role of validation design in both performance evaluation and explanation reliability, particularly in high-stakes domains such as crisis response. Future work will expand this analysis to additional disaster events, explore alternative classifiers (such as transformer-based models), and incorporate cross-lingual or multimodal data to further evaluate model robustness and interpretability.

A. Limitations and Future Work

This study focused on understanding how data splitting strategies influence the interpretability of classification models

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for disaster-related tweets. Although the experiments offer meaningful insights, several limitations should be acknowledged.

First, the analysis was limited to two of the four annotated classification tasks, Informativeness and First Responder Relevance, due to space constraints and clarity of visual explanation. Extending the methodology to include Disaster Type and Secondary Annotation could provide a more complete assessment of feature behavior across different classification goals.

Second, the use of SVM with TF-IDF features provided a transparent and interpretable baseline, but more complex language models (e.g., transformer-based architectures) may capture subtler semantic patterns and could yield different explanations under the same data splitting strategies. Future work could apply this analysis framework to such models, comparing whether the trends observed in feature attribution and rank stability persist.

Additionally, our preprocessing pipeline relied on English stopword removal. Given the international scope of disaster tweets, which often include multilingual content, future work should explore incorporating multilingual stopword sets and tokenization strategies. This could improve both model accuracy and the quality of feature attribution in non-English contexts.

Finally, although SHAP and SVM coefficient analyses were conducted across multiple disasters, our results are still limited by the size and diversity of the underlying dataset. Future studies may benefit from larger multilingual datasets and external validation with real-world deployment cases.

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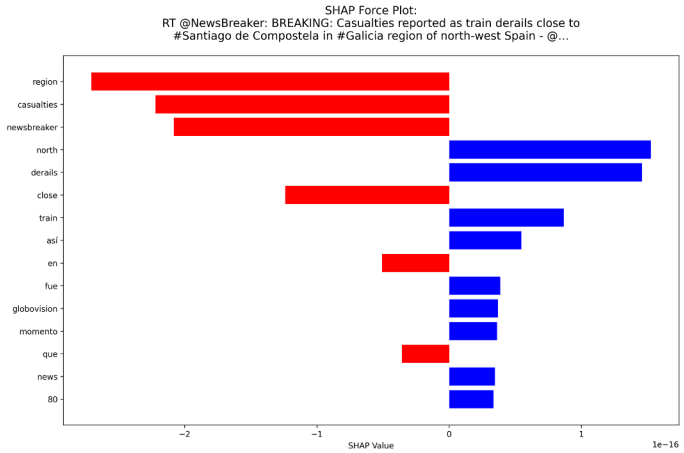
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I. APPENDIX

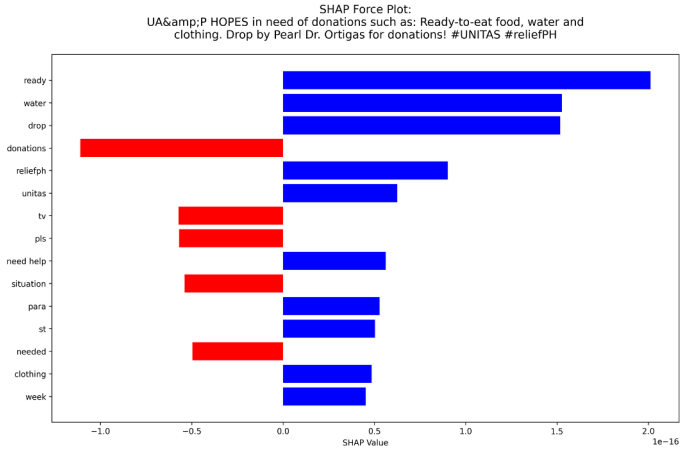
Example SHAP Explanations

To illustrate how SHAP explains individual tweet classifications under the DHV setting, we include two representative examples below. Each case shows the original tweet, its true and predicted labels, model confidence, and the top five SHAP-attributed words influencing the prediction. These examples demonstrate how DHV-trained models highlight contextually relevant terms (e.g., *casualties*, *donations*) that align with correct classification decisions, while suppressing event-specific noise.



Tweet: RT @NewsBreaker: BREAKING: Casualties reported as train derails close to #Santiago de Compostela in #Galicia region of north-west Spain - @…
True Label: Related and informative.
Predicted Label: Related and informative
Confidence: 0.76
Top SHAP Words: casualties (+0.11), breaking (+0.10), newsbreaker (+0.07), reported (+0.06), derails (+0.06)

This example shows that the model correctly prioritized informative disaster terms like *casualties* and *derails*, consistent with accurate prediction.



Tweet: UA&P HOPES in need of donations such as: Ready-to-eat food, water and clothing. Drop by Pearl Dr. Ortigas for donations! #UNITAS #reliefPH
True Label: Related and informative
Predicted Label: Related and informative
Confidence: 0.91
Top SHAP Words: donations (+0.20), water (+0.07), ready (+0.05), rescueph (-0.05), reliefph (+0.05)

Here, the model emphasized high-utility relief terms such as *donations* and *water*, showing appropriate focus on actionable content.

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